t2-w09 — condition C0

Simplification of Algebraic Terms — Simplification of algebraic expressions: order of operations, brackets and word problems

> The curriculum objective alone. No lesson plan, no note.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	absent
Materials	what the lesson was represented with	prompt	absent
Position	where this objective sat in the sequence	prompt	absent
Method	the shape of the teaching, as event labels	prompt	absent
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Simplification of Algebraic Terms

`subtopic`

Simplification of algebraic expressions: order of operations, brackets and word problems

`term`

Second Term

`week`

9

`lesson_objectives`

• carry out order of operation in algebra
• insert/remove bracket and simplify expression
• solve word problems involving algebraic expression

Objective 1 — carry out order of operation in algebra

`objective_text`

carry out order of operation in algebra

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — insert/remove bracket and simplify expression

`objective_text`

insert/remove bracket and simplify expression

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 3 — solve word problems involving algebraic expression

`objective_text`

solve word problems involving algebraic expression

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Using the Order of Operations with Algebraic Expressions

> EXPLANATION
> When you simplify an algebraic expression you must follow the same order of operations you use with numbers:
> 1. **Brackets** – work inside parentheses or brackets first.
> 2. **Exponents** – evaluate any powers.
> 3. **Multiplication and Division** – from left to right.
> 4. **Addition and Subtraction** – from left to right.
>
> Remember that multiplication and division are on the same level, as are addition and subtraction. In algebra the same rules apply, but you also have to keep track of the variables. For example, in the term $3 \times (2x + 4)$ you first simplify inside the brackets, then multiply the result by 3.

> WORKED_EXAMPLE
> **Worked Example**
> Simplify: \(5 \times (2x - 3) + 4 \times x\)
>
> 1. **Brackets first**: Distribute the 5 inside the brackets.
> \(5 \times 2x = 10x\) and \(5 \times (-3) = -15\).
> So the expression becomes \(10x - 15 + 4x\).
> 2. **Combine like terms** (the terms with \(x\)).
> \(10x + 4x = 14x\).
> 3. **Write the final simplified form**.
> \(14x - 15\).
>
> So, \(5(2x-3) + 4x = 14x - 15\).

1. Simplify: \(3 \times (x + 5) - 2x\)
answer: x + 15

2. Simplify: \(4(2y - 1) + 3y\)
answer: 11y - 4

3. Simplify: \(2(3a + 4) - 5a + 6\)
answer: a + 14

Objective 2

Removing Brackets and Simplifying Expressions

EXPLANATION

When you have an algebraic expression with brackets, the first step is to get rid of the brackets. You do this by distributing any number that is outside the brackets to each term inside, and by remembering the sign in front of the brackets.

Key points
- If a plus sign (+) is in front of the brackets, simply write the terms inside the brackets as they are.
- If a minus sign (–) is in front of the brackets, change the sign of every term inside the brackets before removing them.
- After the brackets are removed, combine like terms (terms that have the same variable raised to the same power).

Why it works: The brackets tell us that the operations inside must be done together before anything outside. Distributing the outside number or sign respects the order of operations and lets us simplify the expression step-by-step.

WORKED_EXAMPLE

Example 1
Simplify: $3(2x+5)-4x$

1. Distribute the 3 into the brackets: $3\cdot 2x+3\cdot 5 = 6x+15$
2. Write the expression with the distributed terms: $6x+15-4x$
3. Combine like terms (the x-terms): $6x-4x = 2x$
4. Final simplified form: **$2x+15$**

Example 2
Simplify: $-(4y-3)+2y$

1. The minus sign in front of the brackets changes the sign of each term inside: $-4y+3$
2. Write the expression: $-4y+3+2y$
3. Combine like terms: $-4y+2y = -2y$
4. Final simplified form: **$-2y+3$**

1. Simplify: $5(x-2)+3x$
answer: $8x-10$

2. Simplify: $-(2a+7)+4a-3$
answer: $2a-10$

3. Simplify: $2[3b-(b+4)]$
answer: $4b-8$

Objective 3

Solving Word Problems with Algebraic Expressions

EXPLANATION

When a word problem gives a situation, first translate the words into an algebraic expression. Look for keywords that tell you what to do:
- "total" or "sum" → addition (+)
- "difference" or "less" → subtraction (–)
- "product" or "times" → multiplication (×)
- "quotient" or "divided by" → division (÷)

Next, write the expression using a letter (like \(x\) or \(y\)) for the unknown quantity. If the problem involves several steps, remember the order of operations (brackets, powers, multiplication/division, addition/subtraction). Finally, simplify the expression and, if the problem asks for a value, solve for the unknown.

Key steps:
1. Identify the unknown and assign a letter.
2. Convert each sentence into a mathematical phrase.
3. Combine the phrases into one expression.
4. Use brackets to show which operations happen first.
5. Simplify the expression, applying the order of operations.
6. If required, solve the resulting equation for the unknown.

WORKED_EXAMPLE

Worked Example

Problem: A rectangle has a length that is 3 cm more than its width. If the perimeter of the rectangle is 46 cm, find the width.

Solution:
1. Let the width be \(w\) cm. Then the length is \(w + 3\) cm.
2. The perimeter of a rectangle is \(2\times(\text{length} + \text{width})\).
   So, \(2\big((w + 3) + w\big) = 46\).
3. Simplify inside the brackets: \((w + 3) + w = 2w + 3\).
4. Multiply by 2: \(2(2w + 3) = 46\) → \(4w + 6 = 46\).
5. Subtract 6 from both sides: \(4w = 40\).
6. Divide by 4: \(w = 10\) cm.

Answer: The width is **10 cm**.

1. A garden is rectangular. Its length is twice its width plus 5 metres. If the garden's area is 180 square metres, what is the width?
answer: 10

2. A school bus can hold 40 students. If each student pays a fare of \(x\) dollars and the total fare collected on a trip is \$560, find the fare per student.
answer: 14

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	3	Scope	grounded
Explanation the principle, rule or relationship to review	3	nothing in the input	unsourced
Worked example a problem with its solution worked through	3	nothing in the input	unsourced
	0	nothing in the input	not produced

Figure a drawing the learner reads from		
Practice what the learner does	8 nothing in the input	**unsourced**
Answer what the learner checks against	8 nothing in the input	**unsourced**